

# Supplementary PCS Documentation: Donor Egg IVF Prediction Modelling

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# 1.Overview

## Study Question

This study investigates whether using more complex machine learning models and additional variables could improve prediction ability and model transportability for predicting the chance of having a live birth following donor egg assisted reproductive technology (ART) treatment, also commonly referred to as “IVF”.

## Background

Accurate estimations of IVF success allow patients to make informed decisions on their treatment and set realistic expectations. Currently there are only two publicly available donor egg IVF prediction tools available to patients, the Fitzgerald and Luke model (Fitzgerald et al., 2024; Luke et al., 2014). However, they rely on fewer predictor variables, use traditional statistical models and operate on outdated data. Exploring models of varying complexity, in addition to additional predictor variables, could provide more accurate and personalised estimations as well as robust across different populations.

## Study design

Eight prediction models (linear, tree-based, distance-based, semi-parametric, and neural networks) were developed using 8,380 Australian donor egg embryo transfer cycles with thirteen patient and treatment characteristics. Model performance was benchmarked against the publicly available prediction model, the Fitzgerald model, also known as the donor egg YourIVFSuccess Estimator. The models underwent internal validation, external validation on 1,173 New Zealand treatment cycles and reverse external validation to assess model transportability. Permutation feature importance and accumulated local effects plots assessed the effects of the additional variables across the models.

## PCS Framework

The purpose of this supplementary documentation is to provide transparency and justification on the steps towards the analysis and model development as well as stability analyses towards the developed models. It follows the Predictability, Computability and Stability (PCS) framework for veridical data science (Yu & Kumbier, 2020).

## Model development pipeline

The general pipeline of the model development and evaluation is as follows: data curation, data pre-processing, model training and optimisation, model evaluation, feature importance and interpretation analysis

## 2.Data

### 2.1 Data overview

#### Dataset overview

The data was sourced from the Australia and New Zealand Assisted Reproduction Database (ANZARD), which is a population level clinical quality registry that captures all the data from IVF services across Australia and New Zealand. It captures information about the patient, treatment and outcomes and had been previously used to develop the YourIVFSuccess Estimator, hence making it a suitable dataset to develop and validate clinical prediction models.

#### Connection to the domain problem

ANZARD collects treatment and birth outcome information for all cases where donor egg IVF treatments were performed across Australia and New Zealand. Given that it is a complete coverage dataset, has undergone extensive validation and quality assurance as part of the registry standard and offers a comprehensive range of collected information, it offers a high-quality dataset to develop models on. Furthermore, by using the same registry source as the YourIVFSuccess Estimator, it enables direct benchmarking against the current tool with the newly developed models.

### 2.2 Data Collection

#### Data collection

This study utilises retrospective ANZARD data between 2020-2024, as ANZARD 3.0 variables became available from 2020 onwards and 2024 is the most recent complete year. The study population is restricted to egg recipient females undergoing their first donor egg embryo transfer during this period, so that it allows tracing of live birth outcomes attributed to the initial treatment. The eligibility and exclusion criteria were adapted from the study Oisin et al, which developed the YourIVFSuccess donor egg Estimator, which allows direct benchmarking with the models developed in this study. Thirteen variables, ranging from patient and treatment characteristics, were chosen based on domain knowledge, existing literature, data availability and the chance to further personalise the YourIVFSuccess donor egg Estimator. The Australian cohort was used to develop and internally validate the models, whereas the independent New Zealand cohort was reserved for external validation. Furthermore, reverse external validation was subsequently done by training the models on the New Zealand cohort and evaluating them on the Australian cohort. This study design was chosen as the current YourIVFSuccess egg donor Estimator is developed using Australian data and is intended for the Australian clinical practice, hence making this cohort appropriate as the development dataset.

#### Data limitation

Despite the dataset having a comprehensive population level coverage of donor egg IVF treatments, there are several limitations to the dataset. Since some variables were only collected when the new ANZARD 3.0 version came out in 2020, so the analysis is restricted to data collected from 2020 onwards, which limits the

sample size and follow-up period. Some new variables of interest could not be included because they were only applicable to female-male intending parents which reduces the usefulness across the study cohort and limiting the development of an inclusive model. Furthermore, donor information is limited to donor age only, which limits the ability to determine whether other donor factors could influence live birth outcomes.

### Data accessibility

The ANZARD data is stored on a UNSW managed server in accordance with the registry’s governance and ethical requirements. As the dataset is classified as highly sensitive, it is not publicly available. It is restricted to authorised users with appropriate ethical approval and permission from the ANZARD data custodian.

## 2.3 Data Splitting

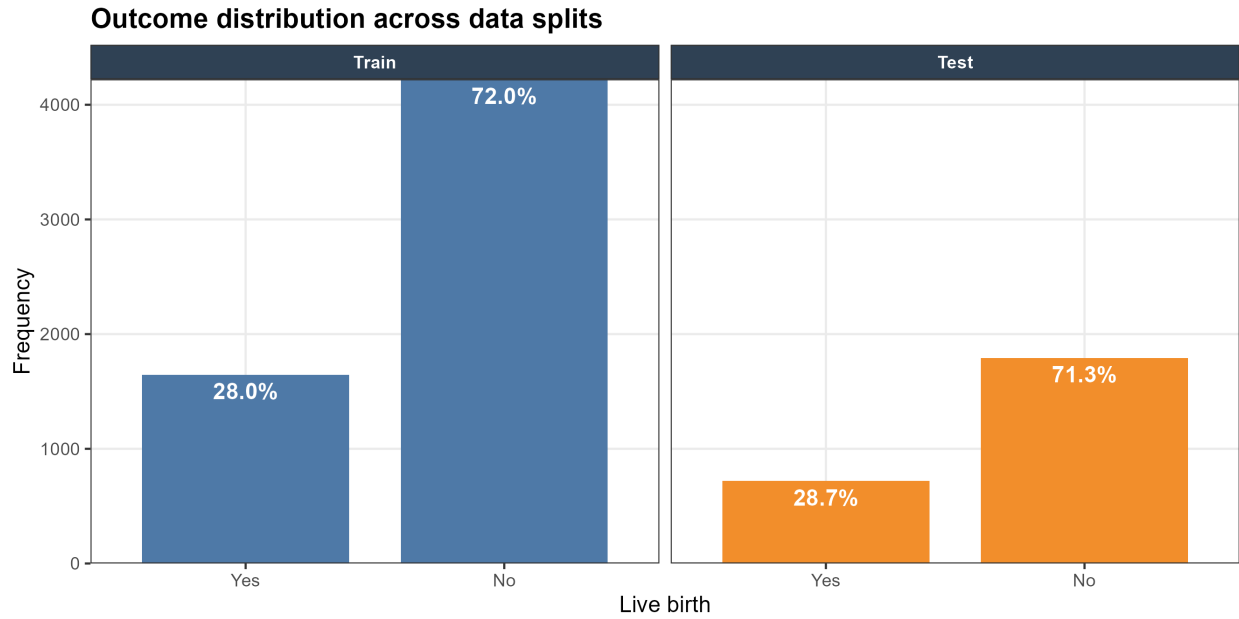
The dataset was split into 70% training and 30% testing sets at a patient level, to ensure all treatments from the same patient was within the same partition. This prevented cycle level leakage, whereby multiple cycles from a single patient would have been split across the two datasets. The split was stratified by live birth and remained unbalanced (at around 30% success rate), to reflect real world probability of donor egg IVF outcomes (**Figure 1-3**).

The 70%/30% ratio split was chosen as it provides sufficient data for model development, whilst having an adequate size test dataset (Bichri et al., 2024; Nguyen et al., 2021). The test dataset was held out throughout the model development such performance can be evaluated in an unbiased manner.

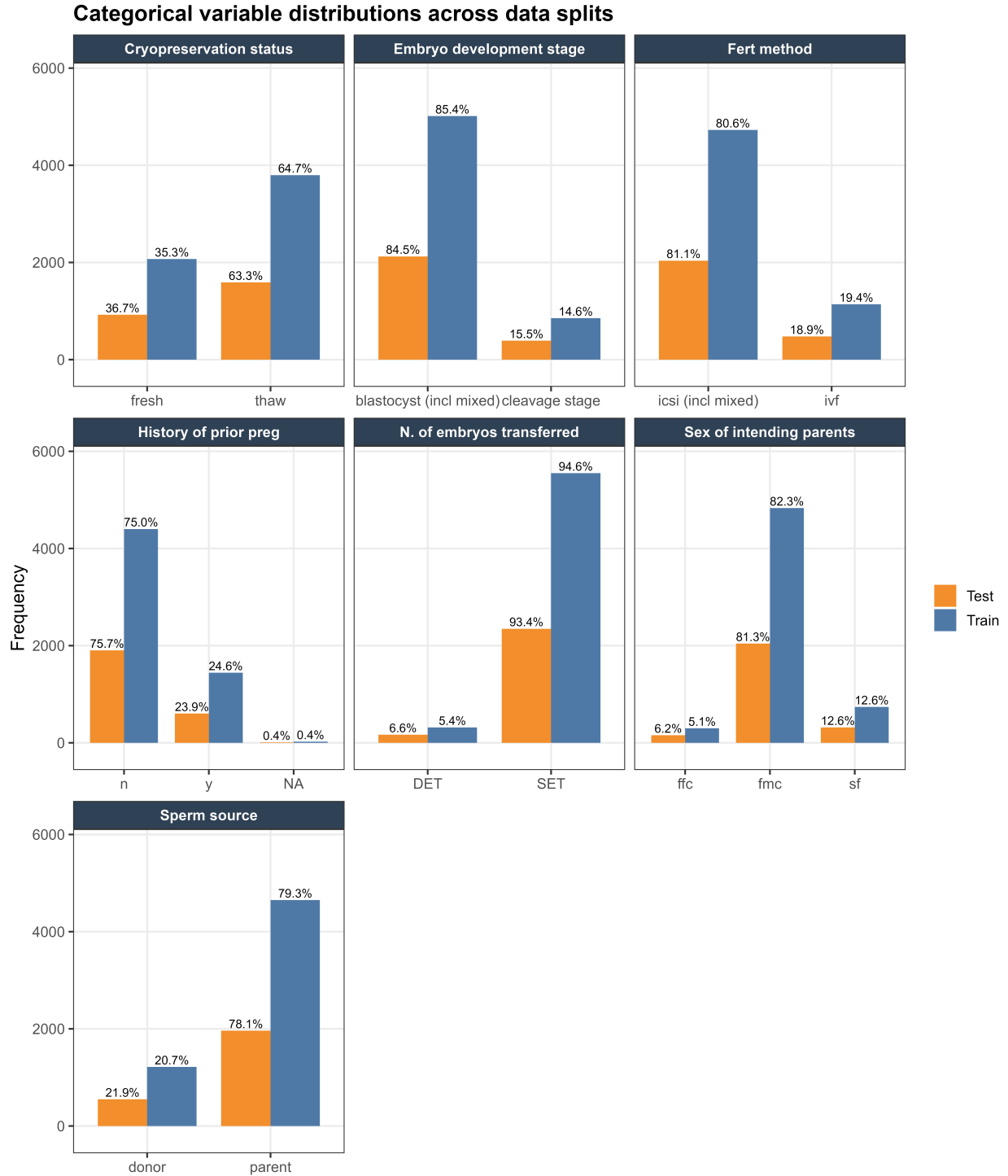
Within the training dataset, repeated stratified 10-fold cross validation (3 repeats) was done to tune the hyperparameters of the models. Hyperparameters were chosen based on the lowest AUC-ROC error, the primary metric used throughout the study which provides a consistent model comparison.

Repeated 10-fold cross validation was selected as it had been shown to reduce variability of performance when compared to non-repeated validation methods (Kim, 2009). Ten-fold was chosen as it is one of the most recommended resampling folds in predictive modelling and in this study showed negligible effects on the predictive performance when compared to the 3 and 5-fold cross validation ( $\pm 0.002$  mean AUC-ROC) and not substantial computational cost (**Figure 4**). The number of repeats was chosen by evaluating the predictive performance of GAM across various commonly used repeat scenarios (repeats of 1, 3, 5, 10). Increasing the number of repeats slightly improved predictive performance though negligible ( $\pm 0.0009$  mean AUC-ROC) despite substantially increasing the computational time (**Figure 5**). Therefore 3 repeats were chosen with the balance of computational efficiency and predictive performance.

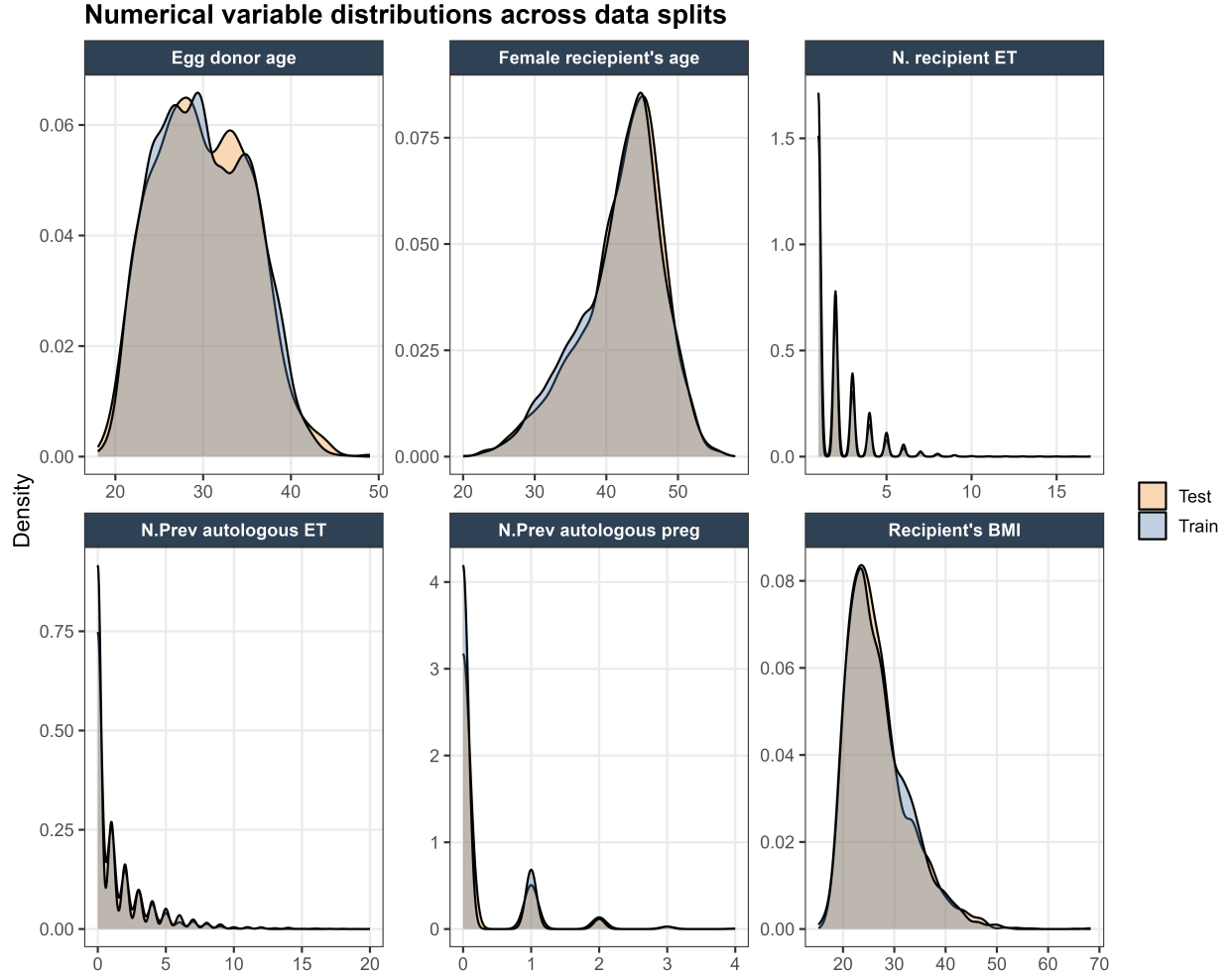




**Figure 1.** Distribution of the live birth outcomes across the 70% training and 30% testing dataset. Bar plots show the frequency of training (blue) and test (orange) following the 70%/30% stratified patient-level split. Percentages displayed represent the proportion within each training or testing dataset.



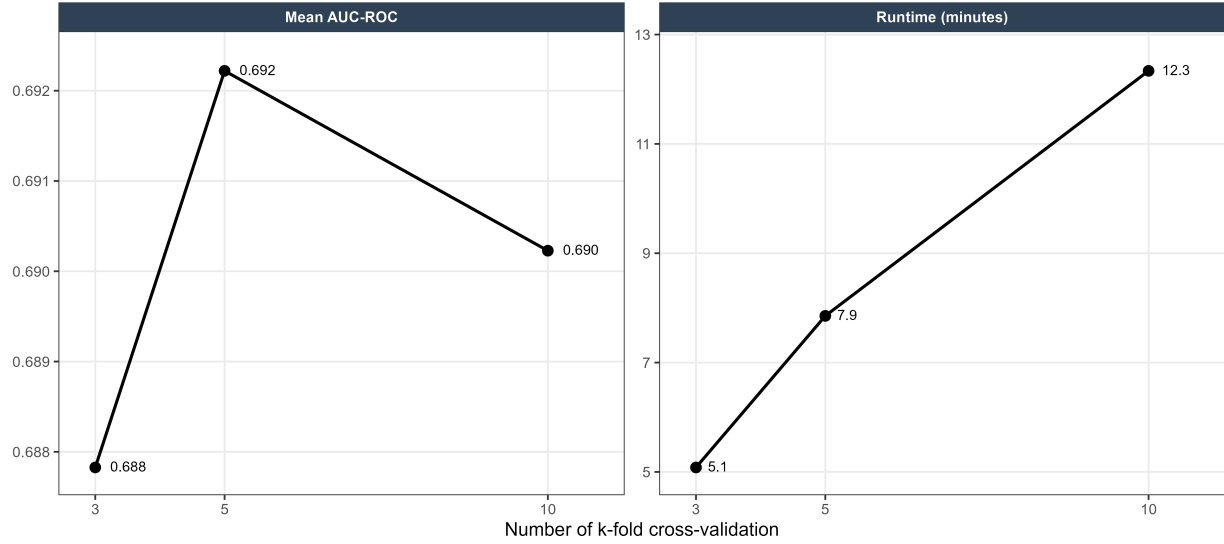
**Figure 2.** Distribution of the categorical predictor variables across the training and testing datasets following the patient-level stratified 70%/30% split. Bar plots show the frequency of each category in the training (blue) and test (orange) dataset. Percentages displayed represent the proportion within each training or testing dataset.



**Figure 3.** Distribution of numerical predictor variables across the training and test datasets following the patient-level stratified 70%/30% split. Kernel density plots compared the distribution across the training (blue) and test (orange) datasets.

### Effect of various k-fold cross-validation on predictive performance and computational cost

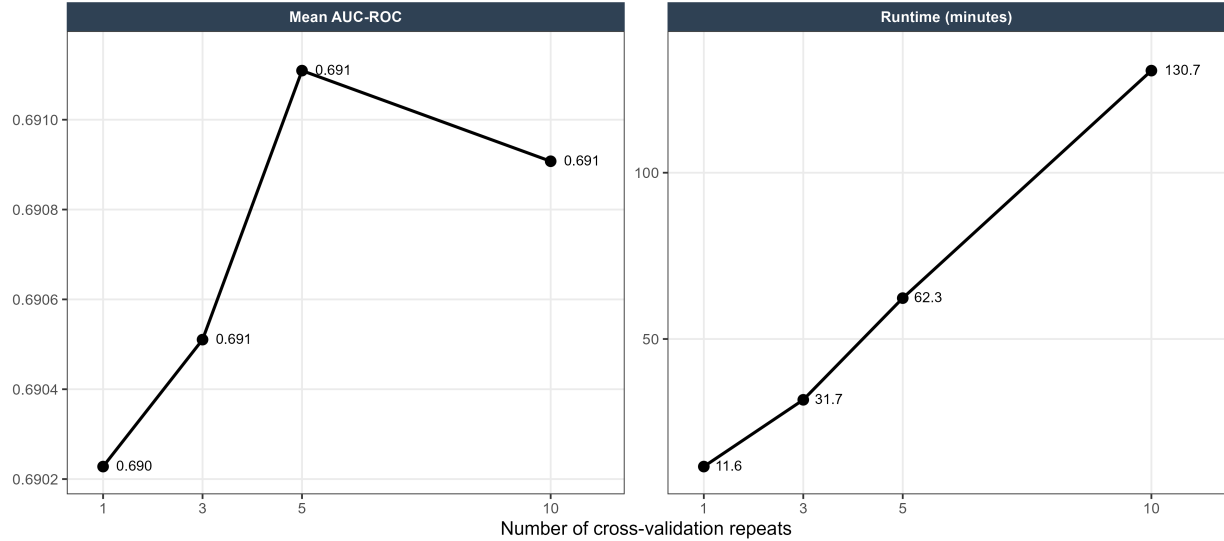
Generalised Additive Model



**Figure 4.** Effect of various k-folds on 1x repeat cross-validation on predictive performance and computational cost using the Generalised Additive Model.

### Effect of repeated cross-validation on predictive performance and computational cost

Generalised Additive Model (10-fold cross-validation)



**Figure 5.** Effect of various repeats on 10-fold cross-validation on predictive performance and computational cost using the Generalised Additive Model.

## 2.4 Data cleaning and preprocessing

### Development environment

Model development was conducted in R.4.5.2 using RStudio. Primary packages included recipes (Kuhn & Wickham, 2024) and caret (Kuhn, 2023) to implement a reproducible processing pipeline and a unified framework for model training, tuning and evaluation. Data.table (Dowle et al., 2019) and dplyr (Wickham et al., 2023) for efficient data manipulation, and ggplot2 (Wickham et al., 2016) was used for its publication

quality visualisations (**Table 1**).

Table 1: Packages used to develop the IVF prediction models using donor eggs, within the R Studio environment. Packages predominantly handled data processing, model development, evaluation and visualisation.

| Package    | Version | Reference               |
|------------|---------|-------------------------|
| data.table | 1.18.4  | Dowle et al. (2019)     |
| dplyr      | 1.2.1   | Wickham et al. (2023)   |
| naniar     | 1.1.0   | Tierney (2023)          |
| rsample    | 1.3.2   | Foster et al. (2023)    |
| recipes    | 1.3.3   | Kuhn & Wickham (2024)   |
| caret      | 7.0-1   | Kuhn (2023)             |
| lubridate  | 1.9.5   | Spinu et al. (2023)     |
| tibble     | 3.3.1   | Wickham et al. (2023)   |
| Metrics    | 0.1.4   | McShane & Albert (2022) |
| PRROC      | 1.4     | Nathan et al. (2021)    |
| iml        | 0.11.4  | Molnar (2022)           |
| ggplot2    | 4.0.3   | Wickham et al. (2016)   |
| plotROC    | 2.3.3   | Sachs (2017)            |
| ggdensity  | 1.0.1   | Otto (2023)             |
| cowplot    | 1.2.0   | Wilke (2024)            |

## Data cleaning and preprocessing

Data cleaning and preprocessing was done to ensure the data was clinically plausible whilst acceptable for the model training.

Distributions were evaluated and values that were out of range were set to missing (**Figure 6-7**). In this case the recipient BMI values ( $<15$  kg/m<sup>2</sup> severely underweight or  $>70$  kg/m<sup>2</sup> morbidly obese), were set to NA. Variables coded as ‘unknown’ using a placeholder value (e.g. age = 99) were also converted to NA, to prevent artificial values influencing the model.

Variables were screen for near zero variance and high pairwise correlation to identify any potential variables that could disproportionately skew the model fitting or obscure other potential predictors (Toloşi & Lengauer, 2011). Spearman’s rank correlation coefficient was chosen as the data presented non-linear relationships amongst the variables (**Figure 8**).

Preprocessing was done independently within the training and test datasets to prevent information leakage. As algorithms require data to be presented in a certain format, preprocessing steps such as transforming categorical variables, standardising continuous variables and imputing missing values are needed.

Categorical variables were converted to one-hot encoded binary variables due to the input requirements of models such as neural networks. Continuous variables were centred and scaled to fit within a unified range, which is a requirement for distance-based and regularised models like K-nearest neighbours (KNN).

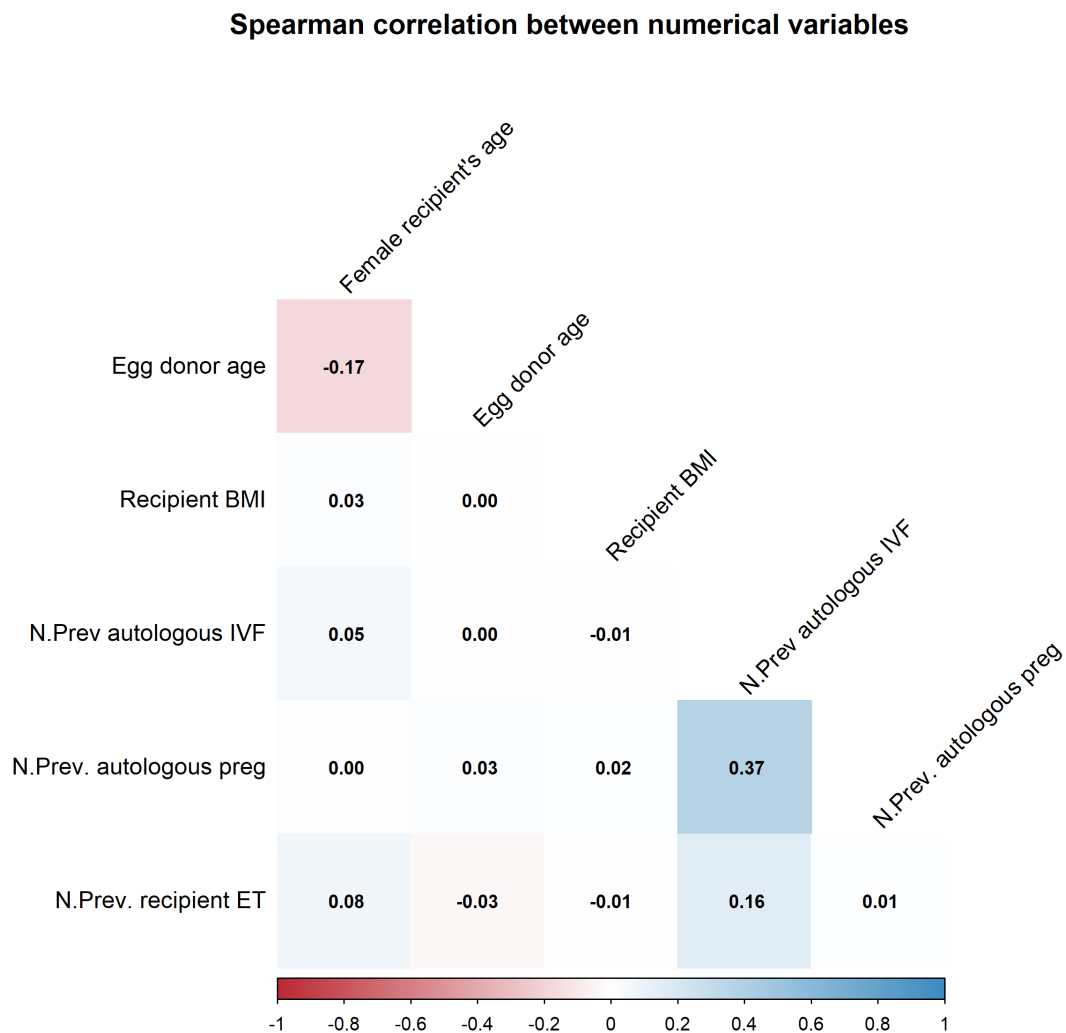
Missing values were imputed using KNN imputation ( $k = 5$ ). Overall, the missingness was low, with recipient BMI having the highest proportion of missing (approximately 6%) (**Figure 9**). Other variables ranged from 0.4-0.8% missing values. As the level of missingness was quite low, there was enough data to make an accurate imputation. Various imputation methods were evaluated to see which imputation created the least AUC-ROC error in a sensitivity analysis, with KNN = 5 proving the lowest imputation error and therefore chosen in the pipeline (**Figure 10**).



**Figure 6.** Pairwise relationship between numerical variables during the model development. The diagonal panel shows the distribution of each variable using kernel density plots, the lower panels display the scatter plots split by live birth (pink) and no live birth (blue) cohorts and the upper quadrant presents the Spearman correlation coefficients for overall cohort (black), live birth (pink) and no live birth (blue). Significant levels indicated as  $P < 0.05$  (\*),  $P < 0.01$  (\*\*) and  $P < 0.001$  (\*\*\*)

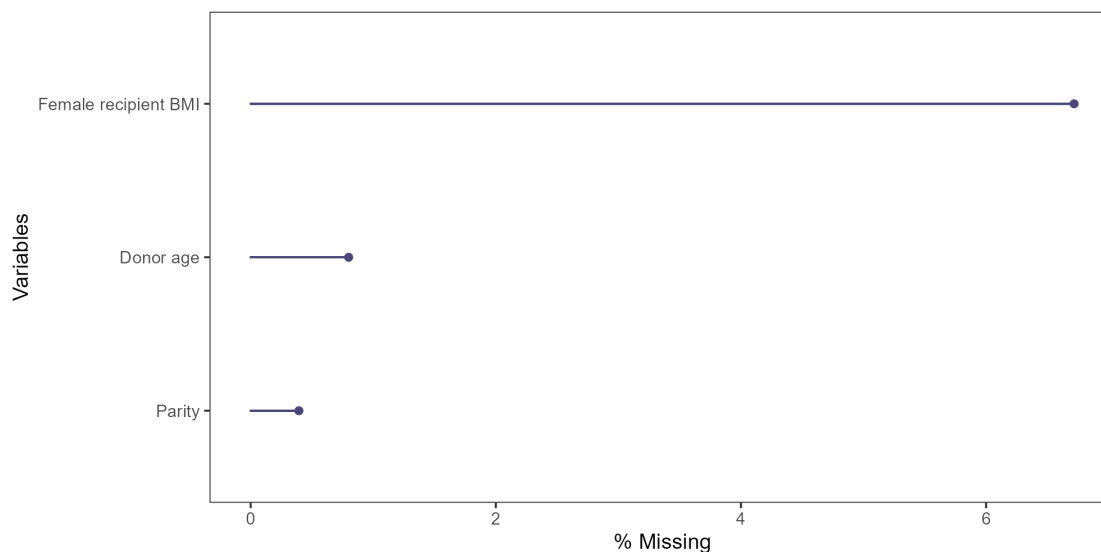


**Figure 7.** Distribution of categorical variables stratified by live birth outcome, live birth = Yes (red) and live birth = No (blue). Percentage displayed on each bar plot represents the proportion of each variable group within the live birth outcome group.

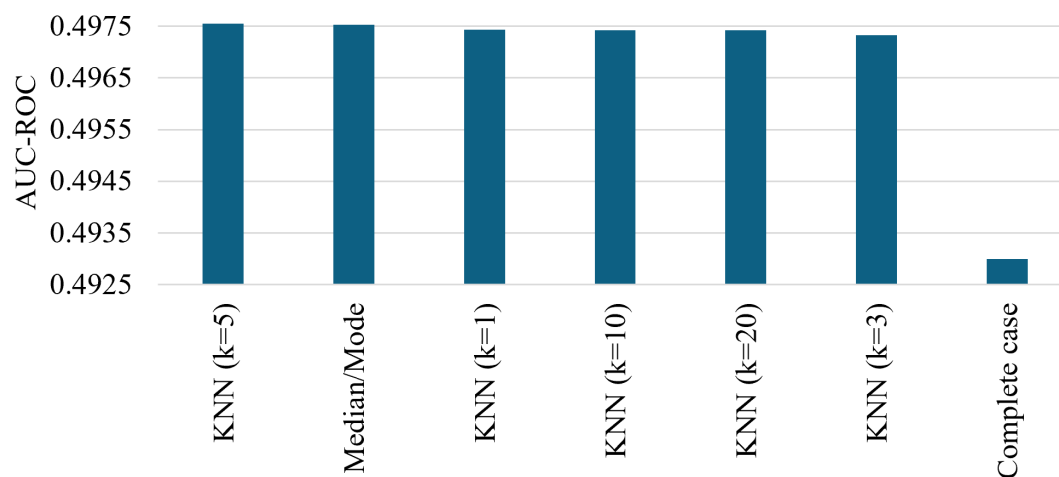


**Figure 8.** Spearman correlation matrix of numerical predictor variables.





**Figure 9.** Proportion (%) of each variable that has missing data points.



**Figure 10.** Comparison of AUC-ROC performance of logistic regression models trained using different missing data handling strategies, including complete case, median/mode imputation and K-nearest neighbours (KNN) imputation on varying neighbour sizes ( $k = 1, 3, 5, 10, 20$ ). Methods are ordered from highest to lowest AUC-ROC performance.

### 3. Prediction modelling

#### 3.1 Modelling methods

The prediction outcome was a live birth outcome after a donor egg embryo transfer, therefore making it appropriate for classification. The goal of the study was to assess whether model complexity influenced prediction outcomes, therefore a diverse range of classification models spanning various levels of complexity and interpretability were chosen. The models included:

- Logistic regression (LR) as it is a widely used approach for classification
- Regularised logistic regression (regLR) as it provides an insight to whether penalisation improves generalisability of LR whilst maintaining interpretability
- Generalised additive models (GAM) as it serves as a baseline model that was used in the YourIVF-Success Estimator and can capture non-linear relationships through splines
- Decision trees (DT) as it is a simple interpretable non-linear tree form model
- Random forest (RF) as it is an ensemble method of decision trees that can handle large complex datasets but is less interpretable
- Extreme gradient boosting (XGBoost) as it is a state-of-the-art algorithm capable of handling complex non-linear relationships and incorporates regularisation however is less interpretable
- Neural networks (NN) as it utilises deep learning networks of non-linear relationships however is less interpretable
- K-nearest neighbours (KNN) as it uses non-parametric distance-based learning methods

The publicly available YourIVFSuccess Estimator was used as a benchmark against the models developed in this study. To ensure a consistent comparison, it was evaluated against the updated data between 2020-2024.

### Evaluation metrics

Three metrics were chosen to assess the models in various ways. AUC-ROC measures a model’s ability to distinguish between the classes. AUC-PR measures how well a model can identify the positive class, which can be particularly useful if the data is imbalanced, which it is in our case, and finally the Brier score which is a joint measure of a model’s calibration and discrimination (Helen R. Sofaer, 2018). All three metrics was taken into consideration, with the top performing model having a well-rounded outcome across the three.

To assess model stability confidence intervals (CI) of 95% was calculated of the model outcomes using the bootstrap method, resampling the test dataset 1000 times. This provides an insight into the variability of model predictive performance in the test cohort through repeated resampling (Johns, 1988).

### 3.2 Prediction check

The eight models were trained using the 70% training dataset following the preprocessing pipeline. A grid search of hyperparameters was tuned during the 3x repeated 10-fold cross validation process, where the hyperparameter was chosen with the lowest AUC-ROC error (**Table 2**).

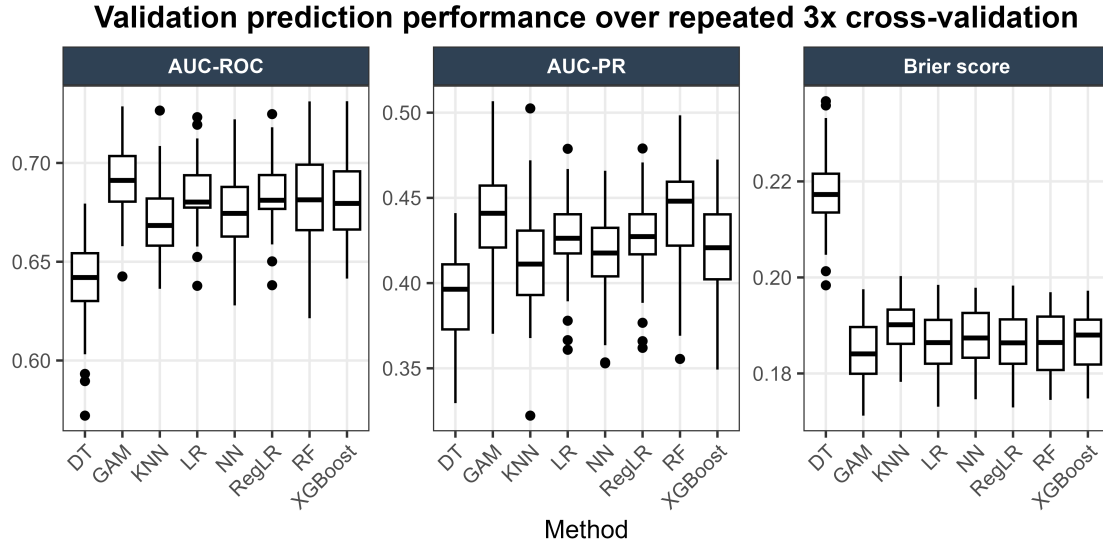
Models were evaluated based on the hold-out method during the cross-validation stage, models that had poor performance in this check and less stable were excluded from further analysis. Confusion matrices were examined to see the balance between sensitivity and specificity (**Table 3**). We can see that GAM had the highest internal performance across the metrics and DT performed the poorest (**Figure 11**).

Table 2: Hyperparameter tuning results and explored parameters for the eight IVF prediction models during the model development component.

| Model          | Caret model | Selected hyperparameters                                                                                       | Hyperparameters explored                                                                                                   |
|----------------|-------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| LR             | glm         | Used all variables in model                                                                                    | NA                                                                                                                         |
| Regularised LR | regLogistic | Cost = 0.6; Loss = L1; Epsilon = 0.001                                                                         | Cost = (0.6, 0.7, 0.8, 0.9, 1); Loss = (L1, L2 dual, L2 primal); Epsilon = (0.01, 0.001, 0.005)                            |
| DT             | rpart       | Cp = 0                                                                                                         | Cp = (0, 0.01, 0.05)                                                                                                       |
| RF             | ranger      | Mtry = 2; Splitrule = gini; Min.node.size = 0.8                                                                | Mtry = (6, 8, 10); Splitrule = (gini, extratrees); Min.node.size = (0.6, 0.8, 1)                                           |
| XGB            | xgbTree     | Nrounds = 200; Max depth = 6; Eta = 0.01; Gamma = 0; Colsample bytree = 1; Min child weight = 1; Subsample = 1 | Nrounds = (100, 200, 500); Max depth = 6; Eta = 0.01; Gamma = 0; Colsample bytree = 1; Min child weight = 1; Subsample = 1 |
| GAM            | gam         | Select = 0.001; Method = GCV.Cp                                                                                | Select = (0.005, 0.01, 0.05); Method = (GCV.Cp, REML)                                                                      |
| NN             | nnet        | Size = 5; Decay = 0.2                                                                                          | Size = (5, 10, 15); Decay = (0, 0.1, 0.2)                                                                                  |
| KNN            | knn         | K = 160                                                                                                        | K = 1 to 200 (increments of 10)                                                                                            |

Table 3: Confusion matrices for the models during training 3x repeated 10-fold cross validation.

| Model   | True Positive | False Positive | False Negative | True Negative |
|---------|---------------|----------------|----------------|---------------|
| LR      | 140           | 176            | 4795           | 12490         |
| RegLR   | 120           | 153            | 4807           | 12513         |
| DT      | 1617          | 2353           | 3318           | 10313         |
| RF      | 136           | 65             | 4799           | 12601         |
| KNN     | 36            | 51             | 4899           | 12651         |
| XGBoost | 357           | 445            | 4578           | 12221         |
| GAM     | 362           | 329            | 4573           | 12337         |
| NN      | 600           | 677            | 4335           | 11989         |



**Figure 11.** Internal validation performance across 3x repeated 10-fold cross validation. Boxplots visualise the distribution across AUC-ROC, AUC-PR and Brier scores for each prediction model across the 30 held-out validation folds.

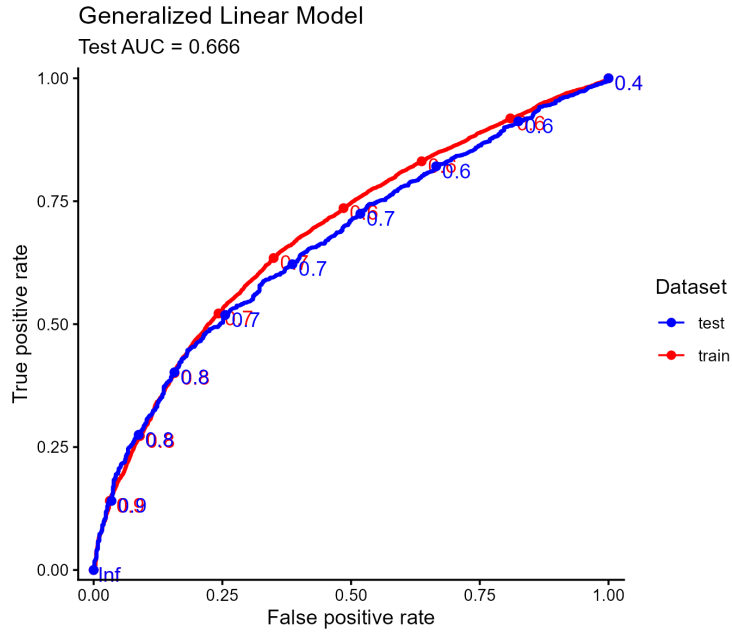
### 3.3 Stability check

Model stability was assessed across several steps to determine whether performance remained consistent across the cross validation of the training dataset through to external validations. During the cross validation, all models except decision tree (DT) demonstrated stable performance across the three evaluation metrics. The DT showed greatest variability and poorer performance, indicating it is sensitive to changes in the training data. Test/train ROC plots indicated that the models predicted consistency across the test/train subsets, indicating low risk of overfitting (**Figures 12-19**).

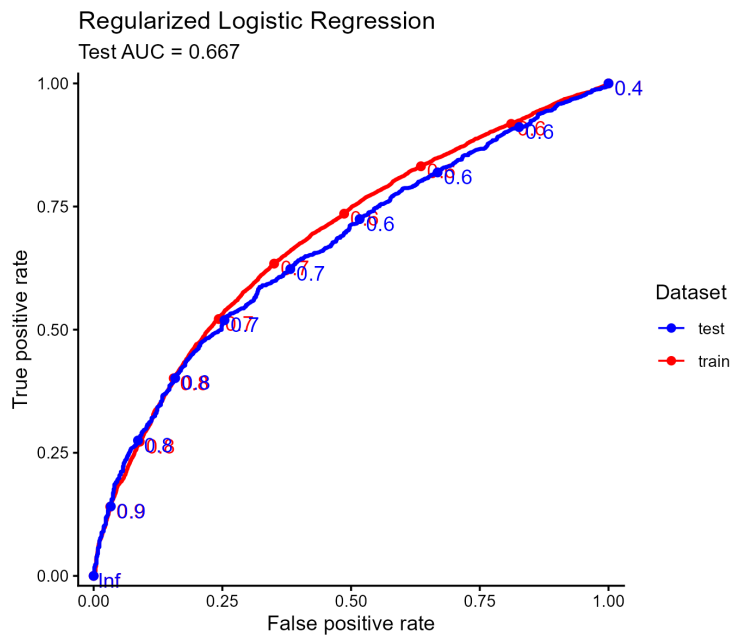
Calibration plots generated from the test dataset were used to inspect the model's prediction probability to the observed live birth outcome. It highlights the confidence of the model's prediction (**Figures 20-27**). All models except DT had good calibration, particularly within the relevant clinical range of 10-40%, whereby 30% is reflected in the real-world scenario.

To quantify the uncertainty of the models during the testing phase, 95% confidence intervals were generated across the three-evaluation metrics using 1000 bootstrap resamples of the test dataset. Narrower confidence intervals indicated greater precision on the performance metrics and increase confidence in the model's prediction. Model stability was assessed through external validation using the New Zealand cohort, detailed more in Chapter 4. Validation Outcomes. Here, performance was generally maintained with only minor metric shifts. Model transportability was further assessed using reverse external validation by training the models on New Zealand data and testing it on the Australian cohort. This helped assess whether the change in metrics was due to population differences rather than model instability.

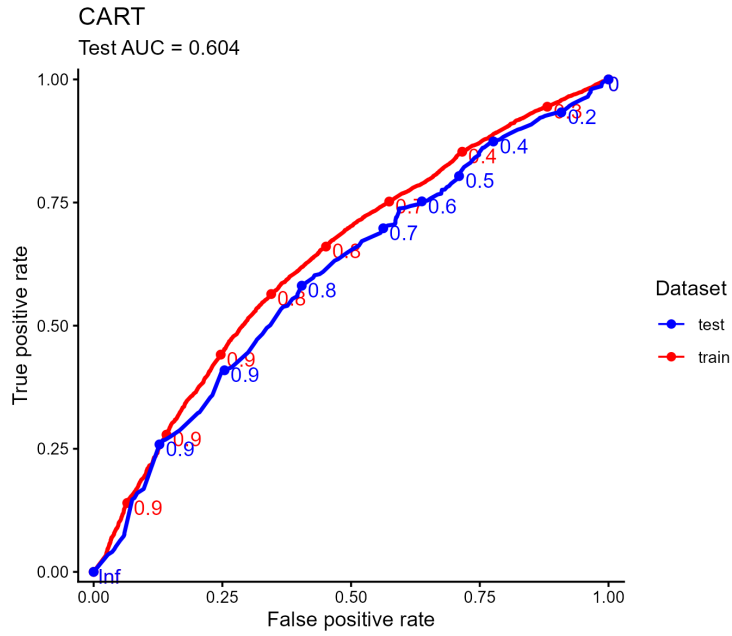
## ROC plots



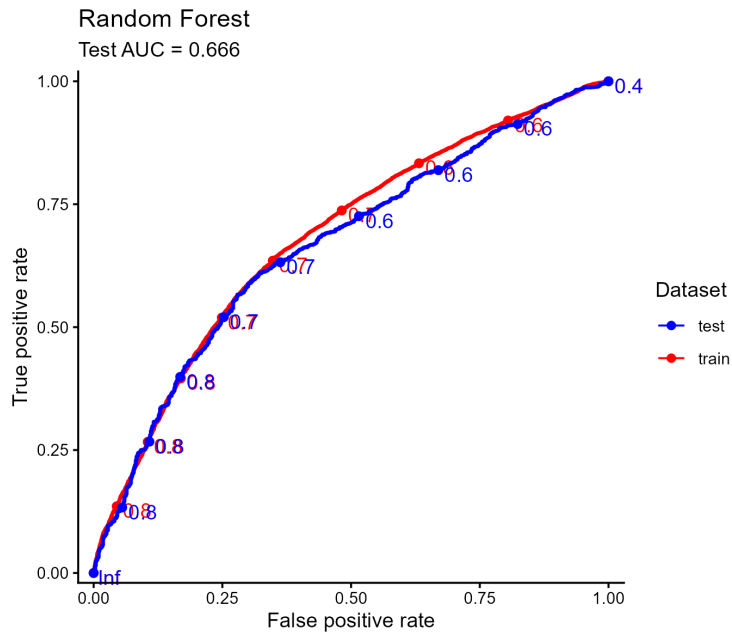
**Figure 12.** Receiving operating curve (ROC) analysis of **LR** during its training and testing process.



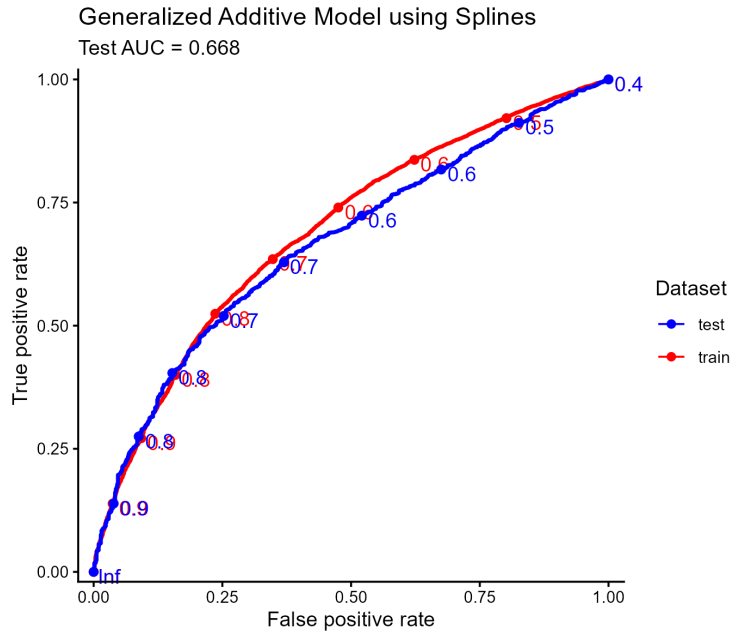
**Figure 13.** Receiving operating curve (ROC) analysis of **regLR** during its training and testing process.



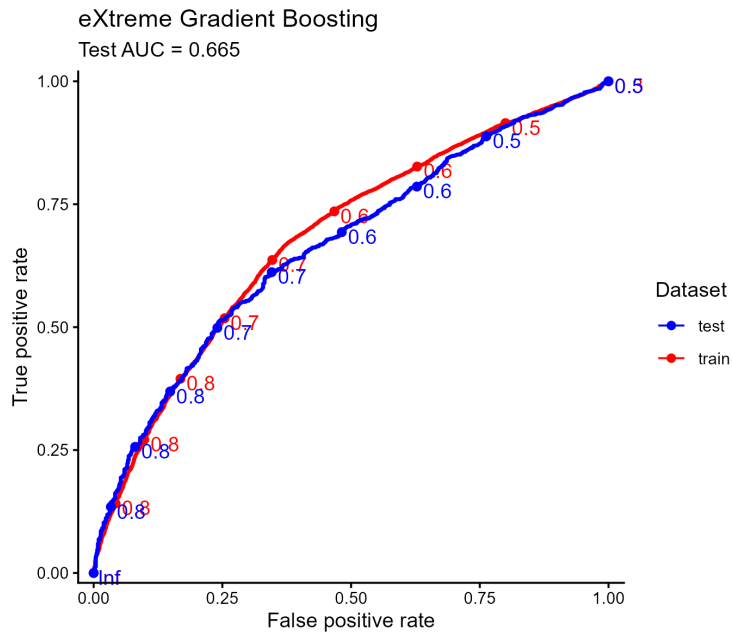
**Figure 14.** Receiving operating curve (ROC) analysis of **DT** during its training and testing process.



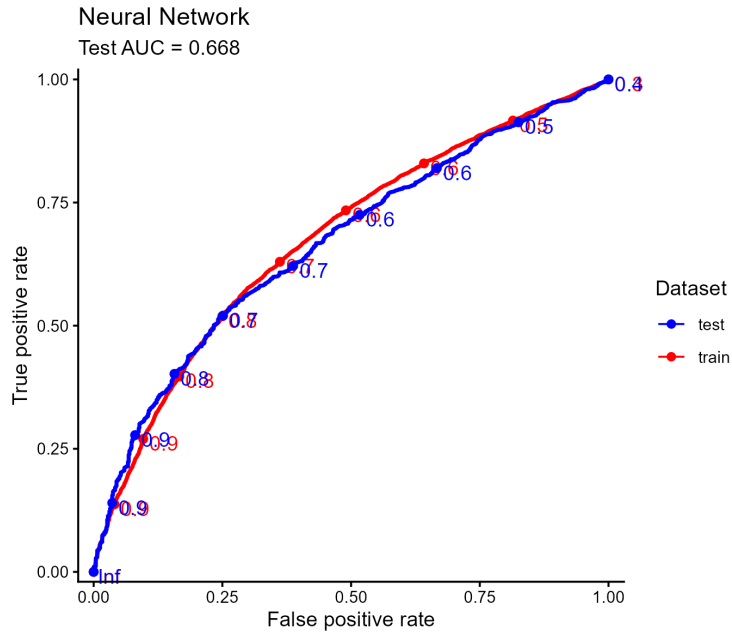
**Figure 15.** Receiving operating curve (ROC) analysis of **RF** during its training and testing process.



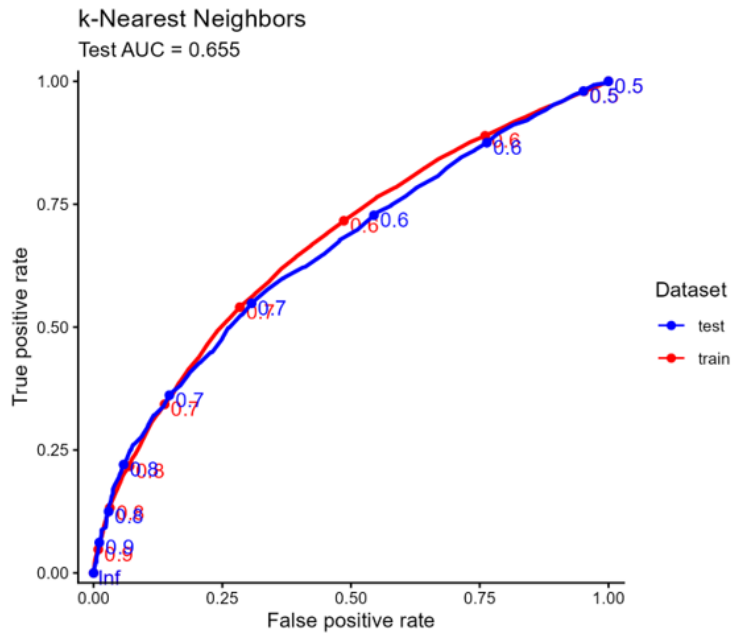
**Figure 16.** Receiving operating curve (ROC) analysis of **GAM** during its training and testing process.



**Figure 17.** Receiving operating curve (ROC) analysis of **XGBoost** during its training and testing process.



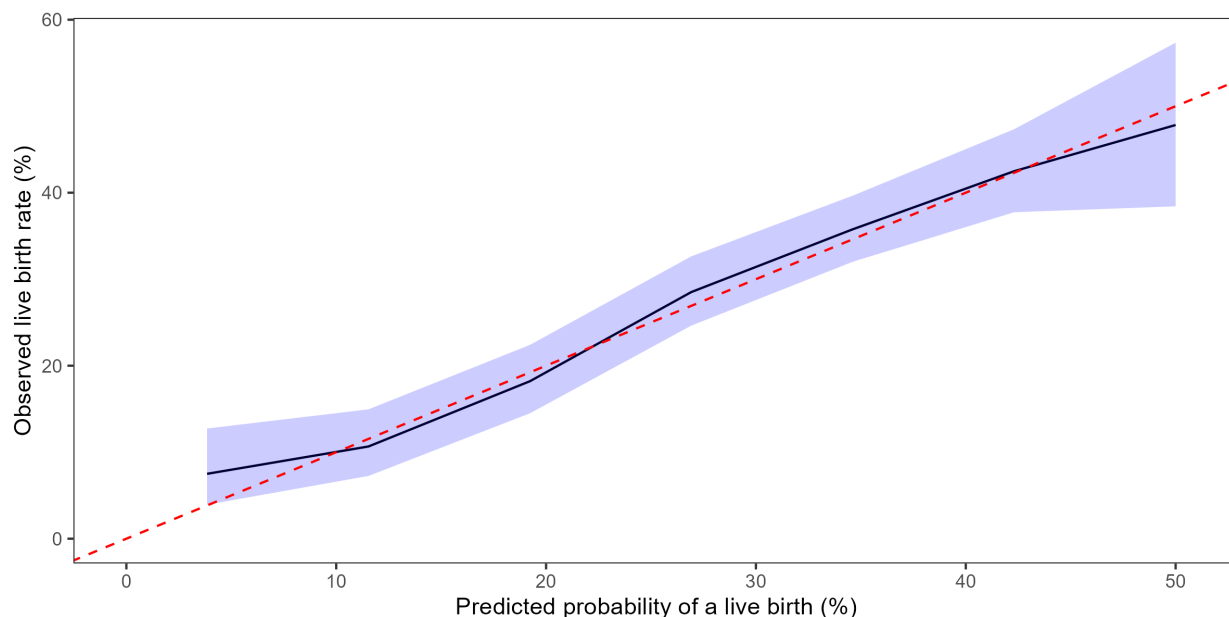
**Figure 18.** Receiving operating curve (ROC) analysis of NN during its training and testing process.



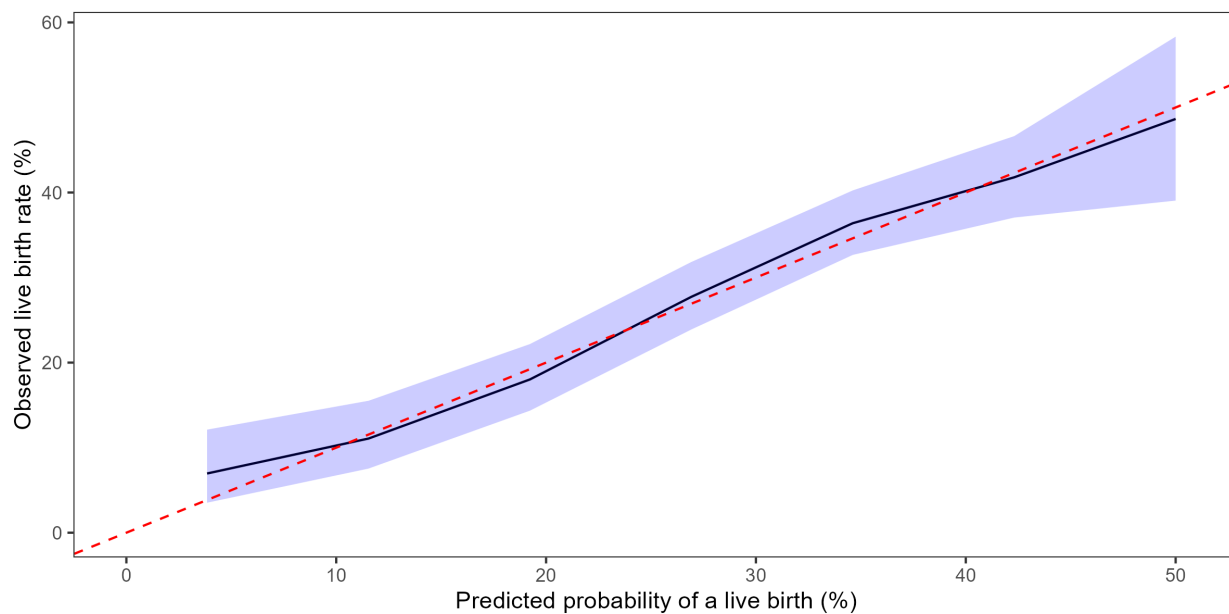
**Figure 19.** Receiving operating curve (ROC) analysis of KNN during its training and testing process.



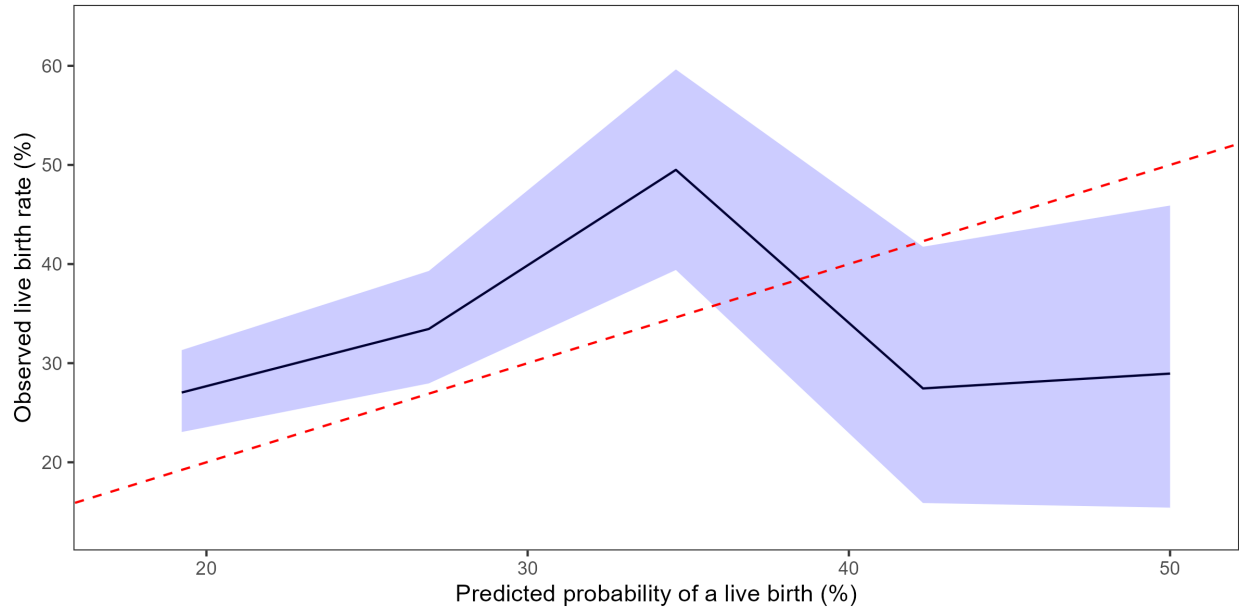
## Calibration plots



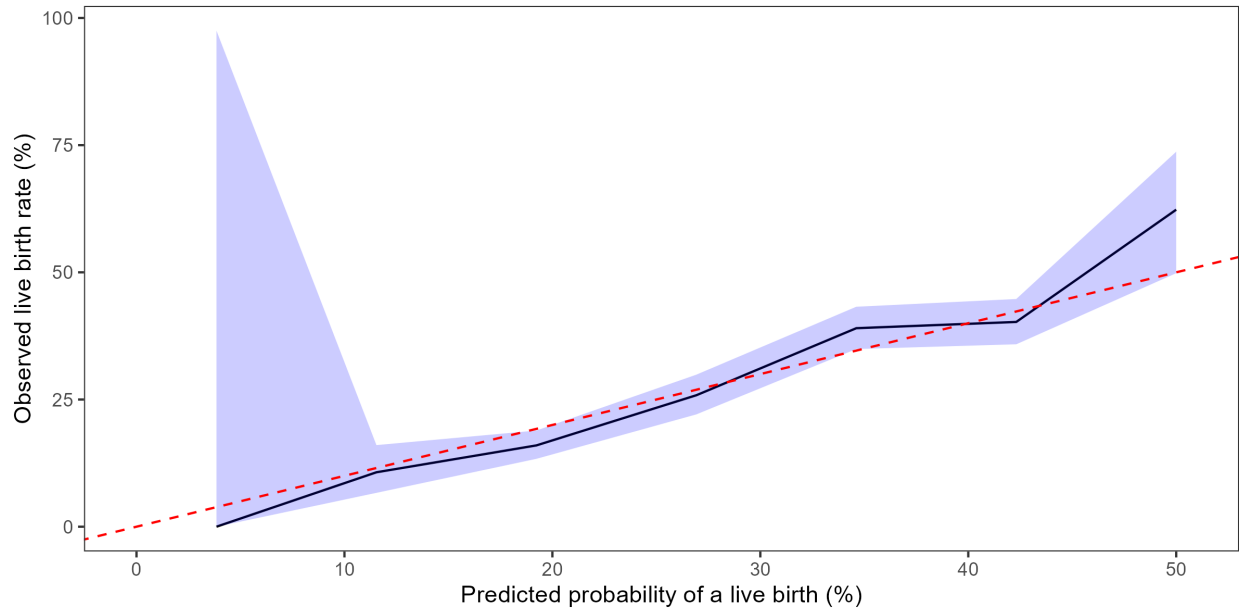
**Figure 20.** Calibration plot of **LR** predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



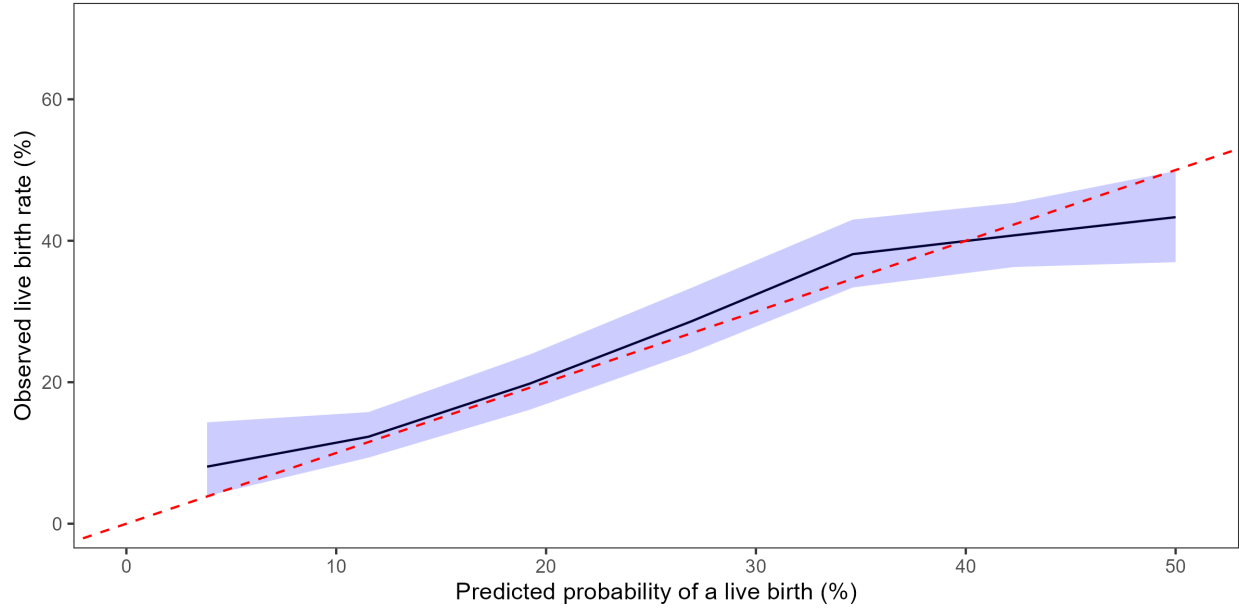
**Figure 21.** Calibration plot of **regLR** predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



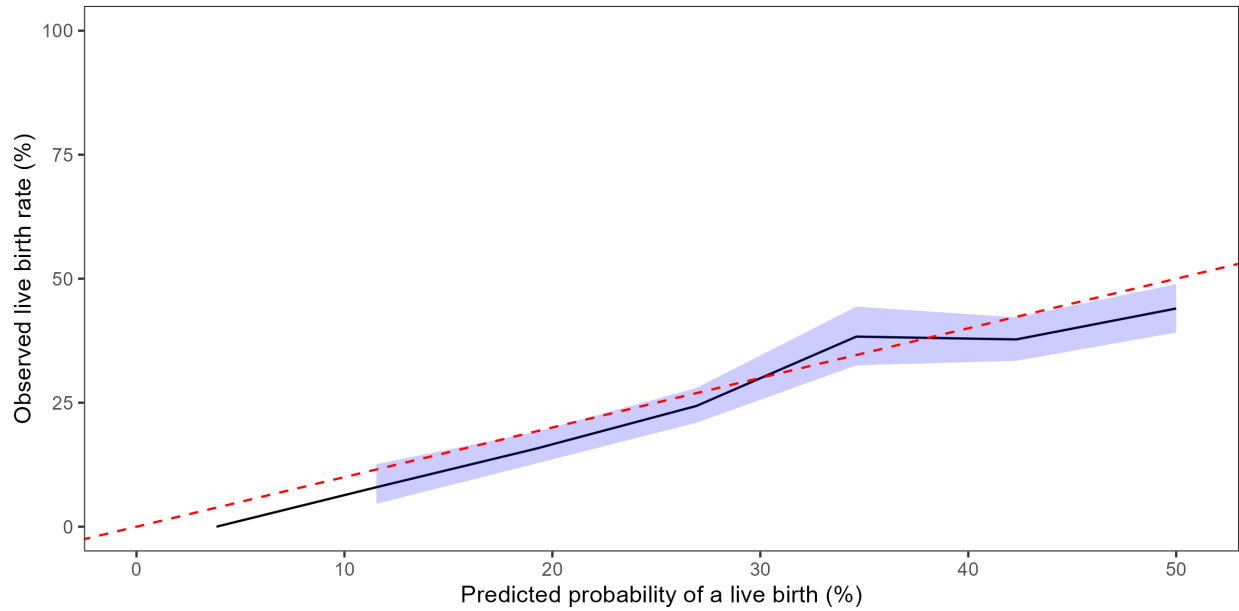
**Figure 22.** Calibration plot of **DT** predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



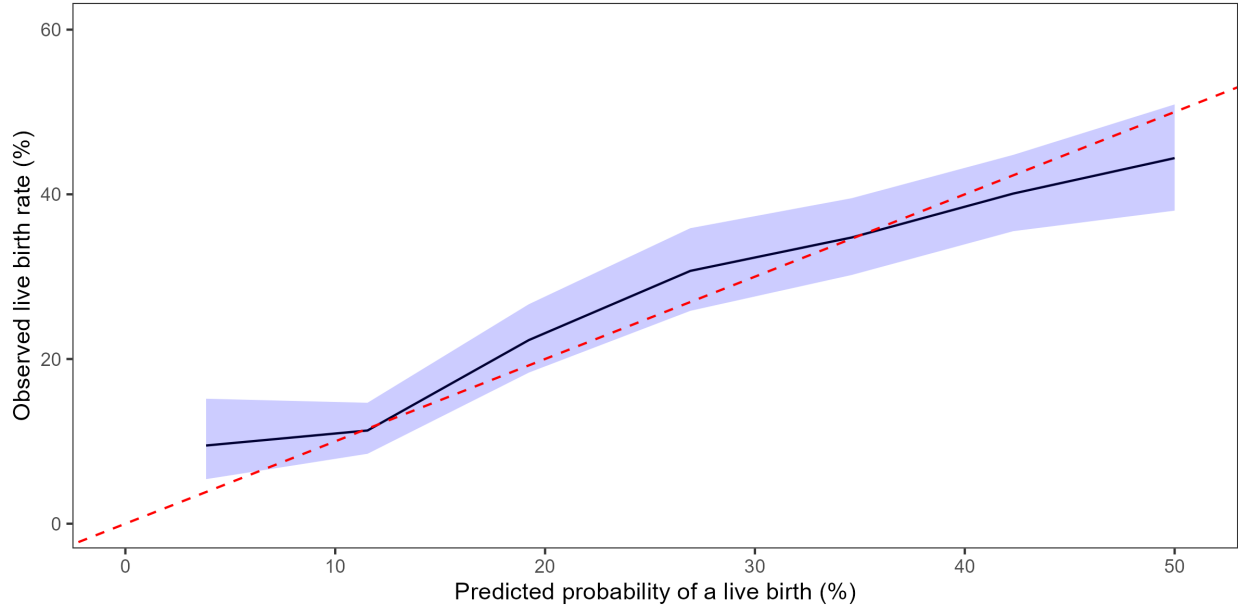
**Figure 23.** Calibration plot of **RF** predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



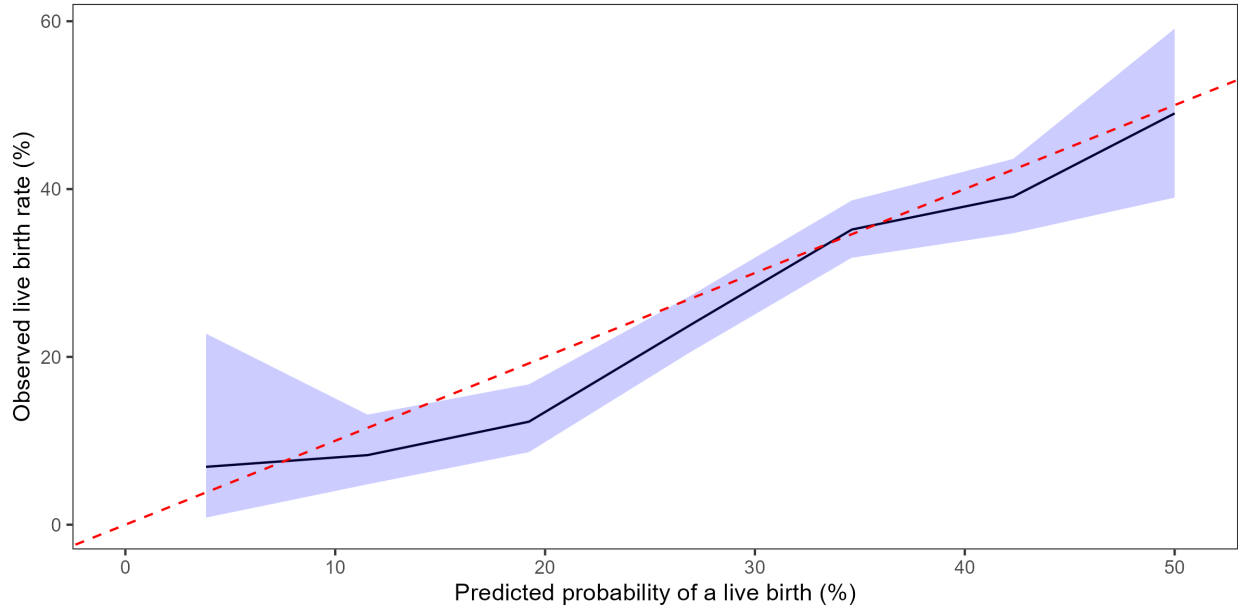
**Figure 24.** Calibration plot of **GAM** predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



**Figure 25.** Calibration plot of **XGBoost** predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



**Figure 26.** Calibration plot of NN predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.



**Figure 27.** Calibration plot of KNN predicting the probability of a live birth using donor eggs. Dotted red line represents optimal calibration where the predicted matches the observed outcome, black line is the model's actual performance and shaded blue is the 95% CI of the models predicted value. CI is based on the binomial-based estimates for each predicted interval.

### 3.4 Interpretability

To better understand the results of the prediction models, interpretable machine learning methods were applied to the models that had adequate prediction and stability performance, which in this case DT was

excluded. Here, the permutation feature importance and accumulated local effects (ALE) was assessed on the test subset to see what was important in the model and how it impacted the prediction.

Permutation feature importance was done to determine which variables were important during the model’s decision making (Breiman, 2001). Importance was determined based on how much the prediction error increases after randomly shuffling the variables one by one. The more important the variable is, the larger the performance degradation. Error was measured based on the change in AUC-ROC, and AUC-ROC was chosen as it was one of the model evaluation metrics. To compare across the models, the feature importance was normalised, using min-max normalisation and averaged across the models. Normalising the numbers between 0 and 1 allows direct comparison across models which have different ranges in results. To get an understanding of the overall importance of the variable, an average was taken across all the models’ AUC-ROC error. Here we could collectively compare which variable was the most important across all models.

ALE plots were done to understand the local effects impact of a variable towards the model predictions. It uncovers certain relationships towards the prediction outcome such as positive/negative or non-linear effects. Histogram plots map the patient distribution on the ALE plots to visualise the distribution of the variable. The top 10 variables from the permutation feature importance analysis were selected for ALE analysis to further interpret the features influence on the model performance whilst maintaining a manageable number of visualisations. It enabled evaluation of the new candidate features to be visualised too.

## 4. Validation outcomes

The test outcomes indicated that GAM achieved the highest AUC-ROC of 0.668, AUC-PR of 0.407 and the lowest Brier score of 0.190, this was followed closely by regularised logistic regression. Increasing model complexity did not meaningfully improve prediction performance over more interpretable models (**Table 4-5**).

When externally evaluating on the New Zealand cohort, GAM also achieved the strongest performance followed by regularised logistic regression (**Table 6**). It achieved higher outcomes. When reverse external validation was done, regularised logistic regression was the strongest followed by GAM (**Table 7**). Such results indicate that interpretable models were strong performing across various scenarios. Variances in overall prediction outcomes are due to the population differences between Australian and New Zealand cohorts.

Overall, the stability analysis indicated that interpretable models like GAM and regLR had consistent performance across the test validation, external validation and reverse external validation (**Table 8**). These findings indicate that these interpretable models provided the best balance of predictive performance, stability and transportability.

Moving onto the interpretability analysis, it was performed on the seven models (DT excluded) that were stable and outperformed the YourIVFSuccess Estimator. Permutation feature importance consistently indicated that the number of recipient embryo transfers was the top feature across the models with an average AUC-ROC % change of -18.7% when the variable was permuted. This was followed by embryo development stage. Importantly, the candidate pre-treatment features of female recipient BMI and sex of intending parents (female-male and female-female) were ranked amongst the top 10 variables across the models, supporting their potential to be included in the future YourIVFSuccess Estimator.

Deep diving into the ALE plots, they were relatively in line with literature on the nature of the features influencing the live birth rate predictions of the models. For instance, as the number of recipient transfers were done, the less likely a live birth would occur, similar to recipient and donor egg age (Hogan et al., 2019; Williams et al., 2022; Zenke & Chetkowski, 2004). Positive predictors include the transfer of a blastocyst stage embryo, a fresh embryo transfer, the recipient having a previous pregnancy. The new variable, particularly female recipient BMI showed inverse relationship with BMI and live birth prediction (Fabozzi et al., 2024; Luke et al., 2014). And reciprocal ivf (female-female couple) showed higher rates of live birth. The

consistency of these relationships across multiple models support the robustness and clinical plausibility of the variables.

#### 4.1 Test confusion matrix

Table 4: Confusion matrices for the final models during the held out test phase.

| Model   | True Positive (TP) | False Positive (FP) | False Negative (FN) | True Negative (TN) |
|---------|--------------------|---------------------|---------------------|--------------------|
| LR      | 10                 | 20                  | 710                 | 1773               |
| RegLR   | 10                 | 16                  | 710                 | 1777               |
| DT      | 220                | 379                 | 500                 | 1414               |
| RF      | 12                 | 16                  | 708                 | 1777               |
| KNN     | 6                  | 8                   | 714                 | 1785               |
| XGBoost | 40                 | 51                  | 680                 | 1742               |
| GAM     | 47                 | 55                  | 673                 | 1738               |
| NN      | 71                 | 81                  | 649                 | 1712               |

#### 4.2 Test prediction outcomes

Table 5: Model prediction performance outcomes (95% CI) of the eight IVF prediction models trained on Australian treatments from the ANZARD dataset (2020-2024).

| Model                                                 | AUC-ROC (95% CI)    | AUC-PR (95% CI)     | Brier score (95% CI) |
|-------------------------------------------------------|---------------------|---------------------|----------------------|
| GAM                                                   | 0.668 (0.646–0.690) | 0.407 (0.375–0.442) | 0.190 (0.182–0.198)  |
| RegLR                                                 | 0.667 (0.646–0.689) | 0.400 (0.370–0.433) | 0.190 (0.183–0.198)  |
| RF                                                    | 0.666 (0.642–0.690) | 0.413 (0.383–0.448) | 0.191 (0.184–0.198)  |
| NN                                                    | 0.668 (0.645–0.690) | 0.401 (0.370–0.432) | 0.191 (0.183–0.199)  |
| LR                                                    | 0.666 (0.643–0.688) | 0.398 (0.365–0.432) | 0.191 (0.184–0.198)  |
| XGB                                                   | 0.665 (0.643–0.687) | 0.404 (0.372–0.439) | 0.192 (0.185–0.199)  |
| KNN                                                   | 0.656 (0.633–0.677) | 0.400 (0.369–0.437) | 0.193 (0.186–0.200)  |
| YourIVFSuccess<br>Estimator tested on<br>updated data | 0.659 (0.637–0.680) | 0.398 (0.366–0.432) | 0.194 (0.187–0.202)  |
| DT                                                    | 0.604 (0.579–0.628) | 0.356 (0.327–0.387) | 0.237 (0.226–0.249)  |

**Note:** Abbreviations: AUC-ROC, area under the receiver operating characteristic curve; AUC-PR, area under the precision–recall curve; CI, confidence interval. Confidence intervals were estimated using 1,000 bootstrap resamples of the test dataset. Models are presented in descending order of overall predictive performance. The best-performing model(s) for each metric are shown in bold.

#### 4.3 External validation outcomes

**Note:** Abbreviations: AUC-ROC, area under the receiver operating characteristic curve; AUC-PR, area under the precision–recall curve; CI, confidence interval. Confidence intervals were estimated using 1,000 bootstrap resamples of the test dataset. Models are presented in descending order of overall predictive performance. The best-performing model(s) for each metric are shown in bold.

Table 6: Model prediction performance outcomes (95% CI) of the eight IVF prediction models externally validated on the New Zealand dataset (2020–2024).

| Model                               | AUC-ROC (95% CI)    | AUC-PR (95% CI)     | Brier score (95% CI) |
|-------------------------------------|---------------------|---------------------|----------------------|
| GAM                                 | 0.671 (0.615–0.734) | 0.441 (0.359–0.548) | 0.200 (0.182–0.217)  |
| LR                                  | 0.671 (0.615–0.732) | 0.432 (0.355–0.531) | 0.202 (0.184–0.218)  |
| RegLR                               | 0.669 (0.608–0.727) | 0.427 (0.350–0.522) | 0.201 (0.184–0.219)  |
| YourIVFSuccess                      | 0.660 (0.598–0.721) | 0.415 (0.334–0.510) | 0.202 (0.182–0.221)  |
| Estimator tested on<br>updated data |                     |                     |                      |
| KNN                                 | 0.645 (0.583–0.709) | 0.402 (0.318–0.493) | 0.204 (0.192–0.217)  |
| RF                                  | 0.625 (0.563–0.687) | 0.395 (0.320–0.494) | 0.208 (0.193–0.224)  |
| NN                                  | 0.623 (0.564–0.685) | 0.378 (0.303–0.459) | 0.227 (0.203–0.250)  |
| XGB                                 | 0.596 (0.533–0.657) | 0.367 (0.294–0.454) | 0.226 (0.206–0.246)  |

Table 7: Reverse external validation performance outcomes (95% CI) of the IVF prediction models trained on the New Zealand dataset and evaluated on the Australian dataset (2020–2024).

| Model | AUC-ROC (95% CI)    | AUC-PR (95% CI)     | Brier score (95% CI) |
|-------|---------------------|---------------------|----------------------|
| RegLR | 0.659 (0.647–0.671) | 0.390 (0.372–0.407) | 0.203 (0.199–0.207)  |
| GAM   | 0.656 (0.643–0.668) | 0.384 (0.367–0.402) | 0.199 (0.195–0.203)  |
| KNN   | 0.654 (0.642–0.666) | 0.386 (0.370–0.404) | 0.199 (0.197–0.202)  |
| RF    | 0.630 (0.617–0.642) | 0.372 (0.355–0.389) | 0.200 (0.197–0.203)  |
| LR    | 0.631 (0.619–0.644) | 0.362 (0.345–0.381) | 0.201 (0.196–0.205)  |
| XGB   | 0.628 (0.615–0.641) | 0.363 (0.347–0.381) | 0.211 (0.207–0.214)  |
| NN    | 0.600 (0.588–0.614) | 0.341 (0.325–0.357) | 0.227 (0.222–0.232)  |

#### 4.4 Reverse external validation outcomes

**Note:** Abbreviations: AUC-ROC, area under the receiver operating characteristic curve; AUC-PR, area under the precision–recall curve; CI, confidence interval. Confidence intervals were estimated using 1,000 bootstrap resamples of the test dataset. Models are presented in descending order of overall predictive performance. The best-performing model(s) for each metric are shown in bold.

#### 4.5 Model stability across validation scenarios

Table 8: Stability of model performance across external validation scenarios

| Model | Mean abs $\Delta$ AUC-ROC |          | Mean abs $\Delta$ AUC-PR |         | Mean abs $\Delta$ Brier |            | Overall<br>Mean<br>Stability<br>Rank |
|-------|---------------------------|----------|--------------------------|---------|-------------------------|------------|--------------------------------------|
|       | AUC-ROC                   | ROC Rank | AUC-PR                   | PR Rank | Brier                   | Brier Rank |                                      |
| RegLR | 0.005                     | 1        | 0.019                    | 2       | 0.008                   | 1          | 1.33                                 |
| KNN   | 0.006                     | 2        | 0.008                    | 1       | 0.012                   | 4          | 2.33                                 |
| GAM   | 0.008                     | 3        | 0.029                    | 3       | 0.010                   | 2          | 2.67                                 |
| RF    | 0.039                     | 6        | 0.030                    | 4       | 0.011                   | 3          | 4.33                                 |
| LR    | 0.020                     | 4        | 0.035                    | 6       | 0.013                   | 5          | 5.00                                 |
| DT    | 0.030                     | 5        | 0.035                    | 5       | 0.019                   | 6          | 5.33                                 |
| XGB   | 0.053                     | 7        | 0.039                    | 7       | 0.027                   | 7          | 7.00                                 |
| NN    | 0.057                     | 8        | 0.042                    | 8       | 0.036                   | 8          | 8.00                                 |

**Note:** Model stability was assessed using the mean absolute change ( $|\Delta|$ ) in each performance metric from

internal validation across the external and reverse external validation scenarios. Smaller mean absolute changes indicate greater stability. Models were ranked separately for AUC-ROC, AUC-PR and Brier score, with rank 1 indicating the greatest stability. The mean stability rank represents the average of the three metric-specific ranks, with lower values indicating greater overall stability.

## 5. Conclusion

To conclude, increasing model complexity did not improve prediction performance and model transportability compared to more interpretable models like GAM. However, by incorporating additional variables, it did improve the model performance over the current YourIVFSuccess Estimator while providing the opportunity to make the YourIVFSuccess Estimator more personalised. Particularly, recipient BMI and sex of the intending parents emerged as promising variables. Although these findings are based on some newly collected variables, further validation would be beneficial once more data becomes available. Overall, the study demonstrates that interpretable transparent models can provide a robust and trustworthy clinical decision aid for donor egg IVF.

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